Predicting SpaceX Falcon 9 First-Stage Landing Success

Geethima Rajapaksha

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OUTLINE



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EXECUTIVE SUMMARY

- **Goal:**Predict SpaceX Falcon 9 first-stage landing success to estimate launch costs accuratetly.
- **Key Results**: Landing success increased to ~85% by 2019. KSC LC-39A had the highest success rate (41.7%).
- **Best ML Model**: All models achieved 83.33% accuracy; Logistic Regression / SVM are recommended for low complexity.

- **Methodology:**Collected data via SpaceX REST API & Web Scraping, performed EDA (Pandas, SQL, Seaborn), built interactive maps (Folium) & dashboards (Plotly Dash), and trained 4 classification models.

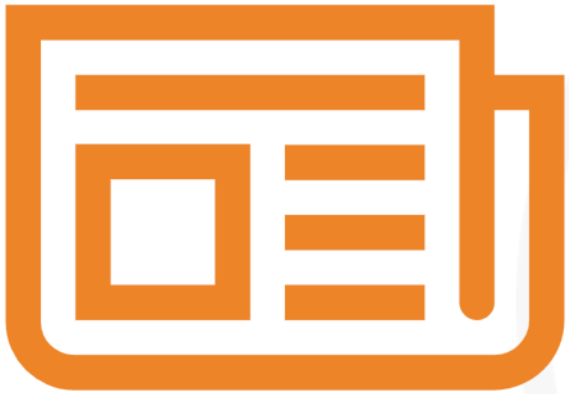
https://github.com/Geethima-Rajapaksha/ibm-datascience-capstone-project

INTRODUCTION



- SpaceX saves money by reusing the Falcon 9 first stage.
- Goal: predict whether the first stage will land successfully.
- Key Questions:
  - Which factors affect landing success?
  - Has the success rate improved over time?
  - Which model predicts best?

METHODOLOGY

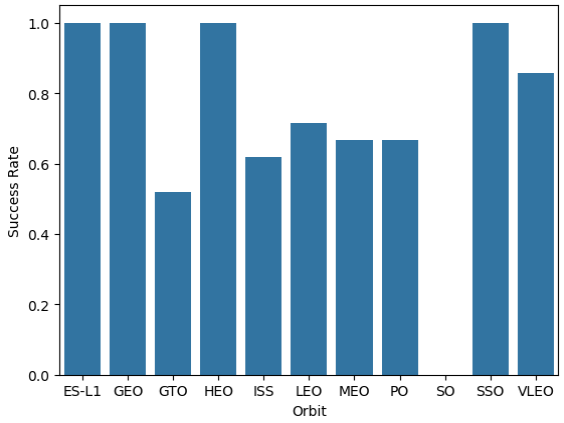


- Data collection
  - SpaceX REST API (requests + json_normalize)
  - Filtered to Falcon 9 launches
- Data Wrangling
  - Handled missing values
  - Created the class label(1=landed, 0=did not land)
- Exploratory data analysis (EDA)
  - Visualizations(Matplotlib/seaborn) and SQL queries
- Interactive visual analytics
  - Folium map and Plotly Dash dashboard
- Predictive analysis (classification)
  - Standarized data, train/test split
  - Tuned Logistic Regression, SVM, Decision Tree and KNN with GridSearchCV

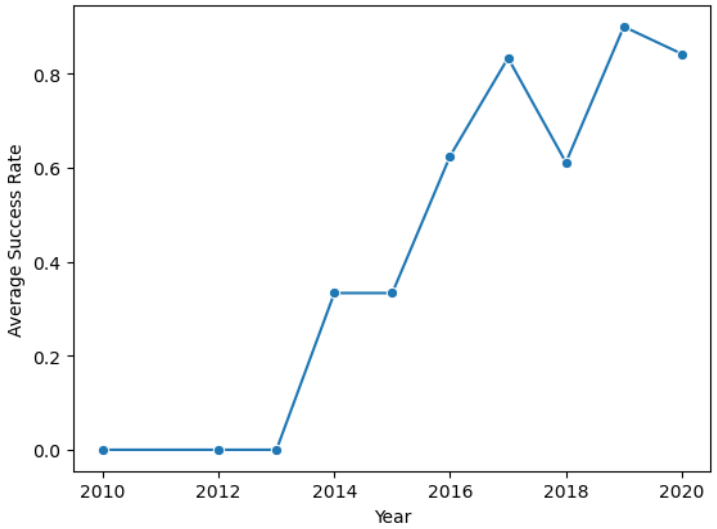
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RESULTS

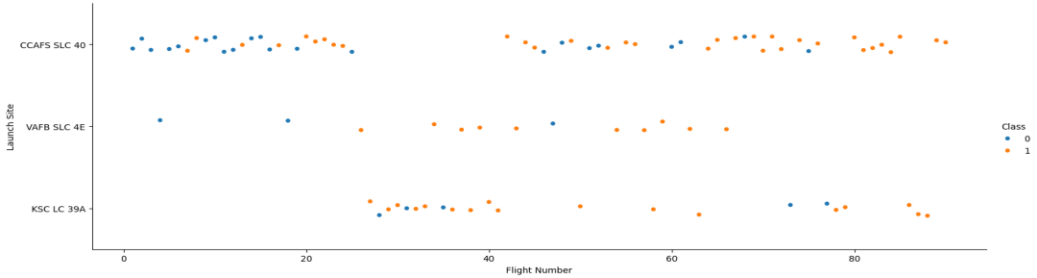
Success rate by orbit



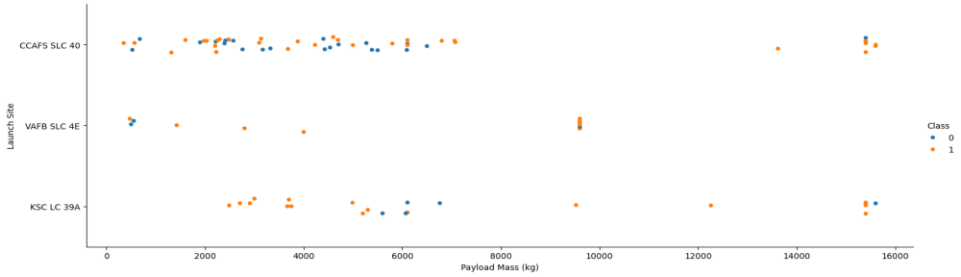
Success rate by year



Flight Number vs Launch Site



Payload Mass vs Launch Site



https://github.com/Geethima-Rajapaksha/ibm-datascience-capstone-project/tree/Exploring-and-Preparing-data

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EDA WITH SQL

Query 1:

SpaceX uses 4 launch sites

Launch_Site

CCAFS LC-40

VAFB SLC-4E

KSC LC-39A

CCAFS SLC-40

Query 2:

Landing outcomes

```
%%sql
SELECT Landing_Outcome, COUNT(*) AS outcome_count
FROM SPACEXTABLE
WHERE Date BETWEEN '2010-06-04' AND '2017-03-20'
GROUP BY Landing_Outcome
ORDER BY outcome_count DESC;
```

https://github.com/Geethima-Rajapaksha/ibm-datascience-capstone-project/tree/Exploring-and-Preparing-data

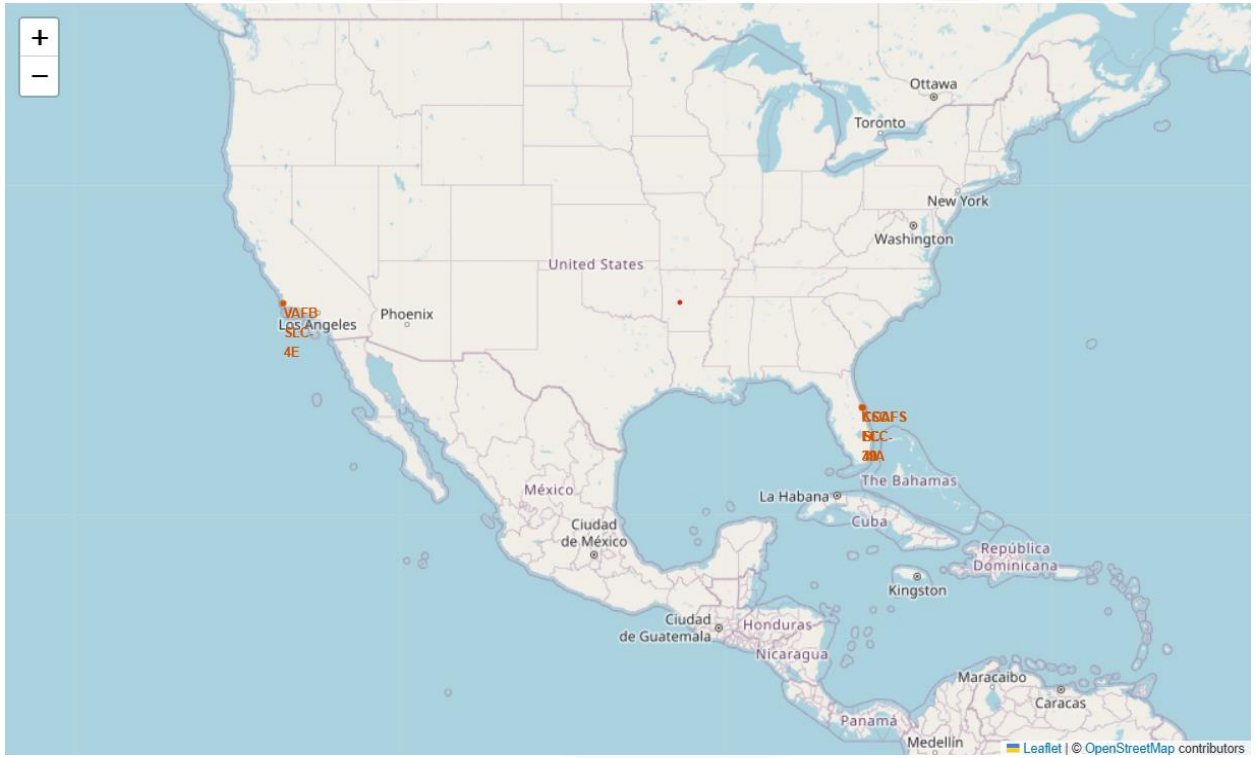
INTERACTIVE MAP WITH FOLIUM

Findings

- All 4 launch sites are on the US coast
- 1 is in California and 3 are in Florida

Implications

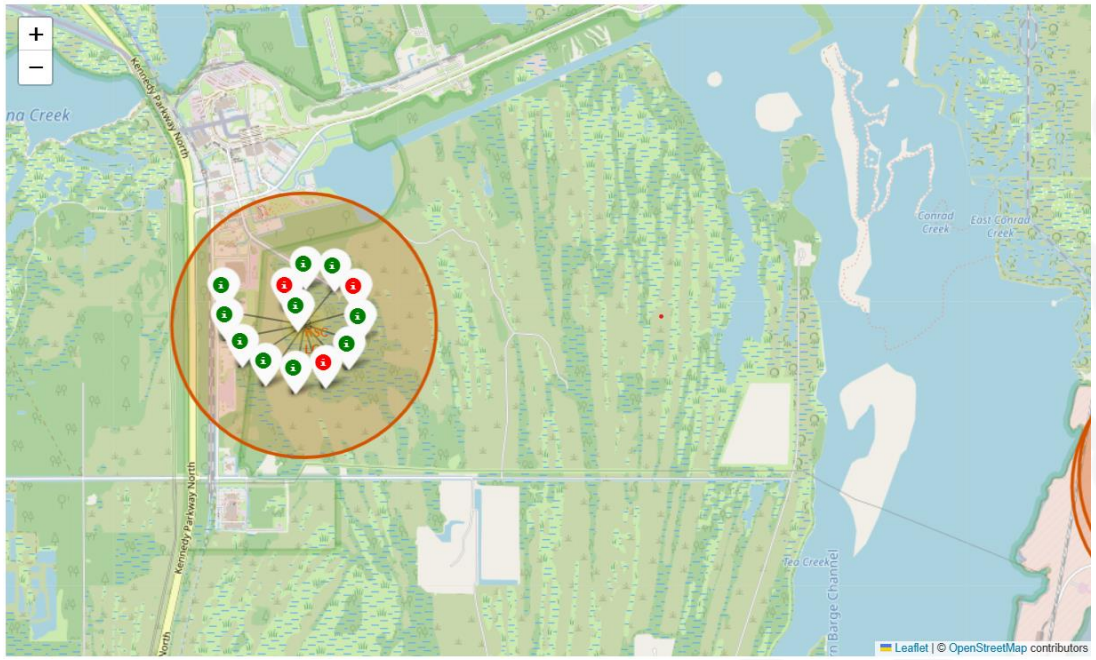
- Launching over the ocean keeps debris away from populated areas.
- Southern sites get a speed boost from Earth's rotation



https://github.com/Geethima-Rajapaksha/ibm-datascience-capstone-project/tree/Folium-Map

LAUNCH SUCCESS BY SITE(FOLIUM)

- KSCLC-39A has 13 launches; CCAFS sites have 33. Green = success, red = failure.





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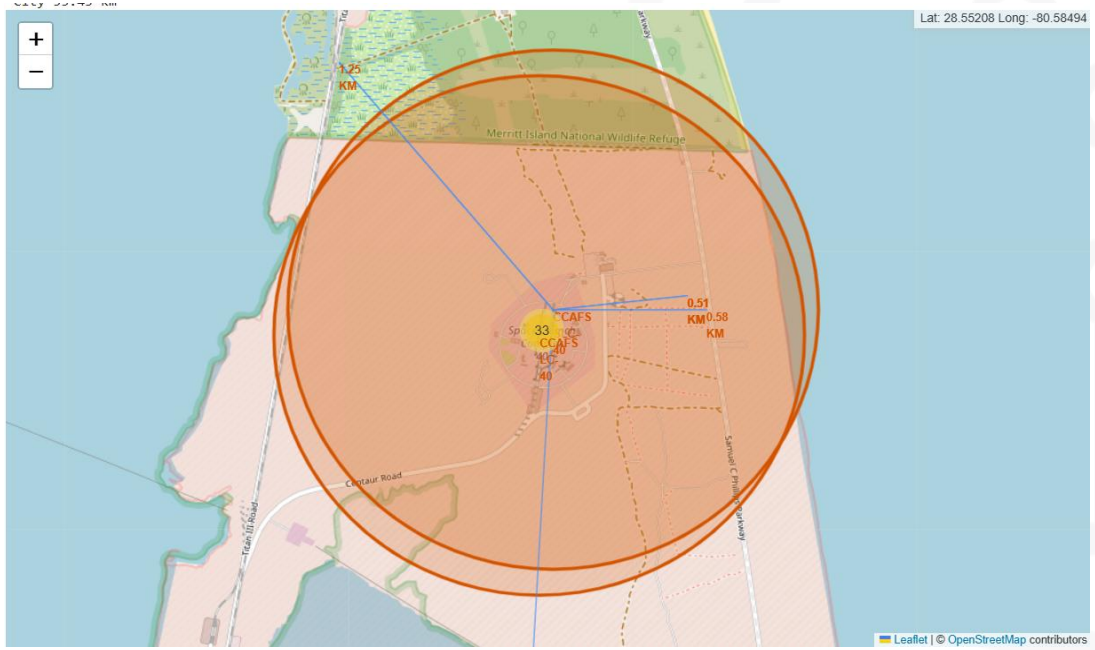
LAUNCH SITE PROXIMITIES(FOLIUM)

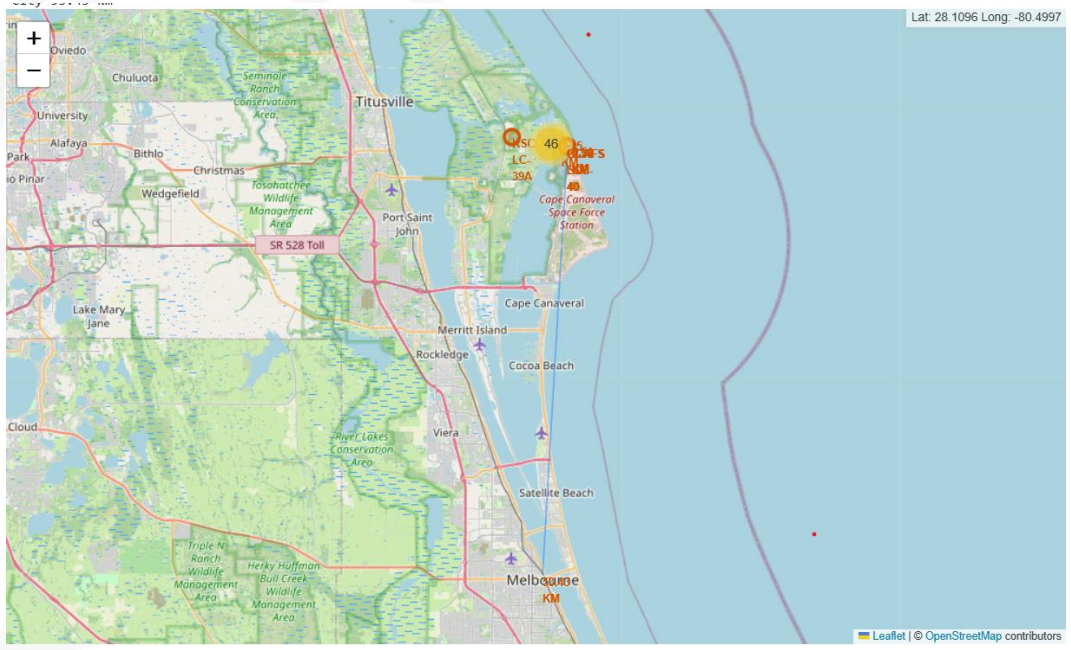
Findings

- CCAFS is about 0.5km from the coast
- It is close to a hightway (0.58km) and a railway (1.25km)
- The nearest city is about 53 km away

Implications

- Coastal sites let rockets launch over the ocean, away from people
- Roads and rails make it easy to move rockets and equipment
- Distance from cities limits the risk if a launch fails



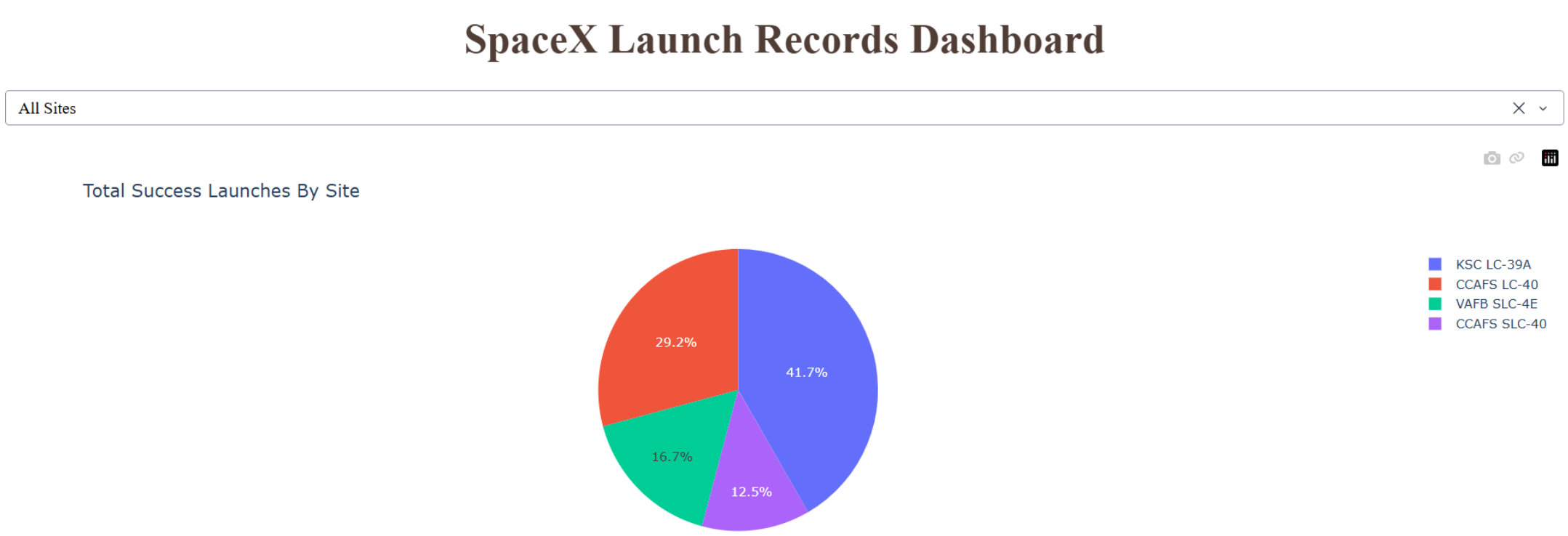


https://github.com/Geethima-Rajapaksha/ibm-datascience-capstone-project/tree/Folium-Map

DASHBOARD

https://github.com/Geethima-Rajapaksha/ibm-datascience-capstone-project/tree/Build-an-Interactive-Dashboard-with-Plotly-Dash

DASHBOARD TAB 1



https://github.com/Geethima-Rajapaksha/ibm-datascience-capstone-project/tree/Build-an-Interactive-Dashboard-with-Plotly-Dash

DASHBOARD TAB 2



https://github.com/Geethima-Rajapaksha/ibm-datascience-capstone-project/tree/Build-an-Interactive-Dashboard-with-Plotly-Dash

DISCUSSION/ FINDINGS

Finding

- **KSC LC-39A Leader:** KSC LC-39A had the highest proportion of successful launches at **41.7%**.
- **Success Trajectory:** Landing success rate increased significantly from **0%** (2010–2013) to **~85%** by 2019.
- **Geographic Strategic Location:** All launch sites are located on US coasts, situated **~0.5 km** from the nearest city (CCAFS).

Implications:

- Safety Protocol: Coastal launch sites direct debris over the ocean, minimizing risk to populated areas.
- Cost Advantage: Reusable boosters dramatically reduce spaceflight costs ($62M SpaceX vs $165M competitor baseline).
- Key Factors: Launch site, payload mass, and booster version are essential predictive variables for landing success.

DATA PREPARATION AND PIPELINE

Machine Learning

Data Preprocessing

- Created binary target vector y from column Class (Landing Success = 1, Failure = 0).
- Created feature matrix X with One-Hot Encoded categorical variables.
- *Code line:* y = df['Class'].to_numpy()

Standardization

- Scaled features using StandardScaler() to ensure zero mean and unit variance across features.
- *Code line:* X = StandardScaler().fit_transform(X)

Train/Test Split & GridSearch

- Split data into training and test sets using train_test_split with an **80/20 ratio** (random_state = 2).
- Used GridSearchCV with **10-fold cross-validation** for hyperparameter tuning.

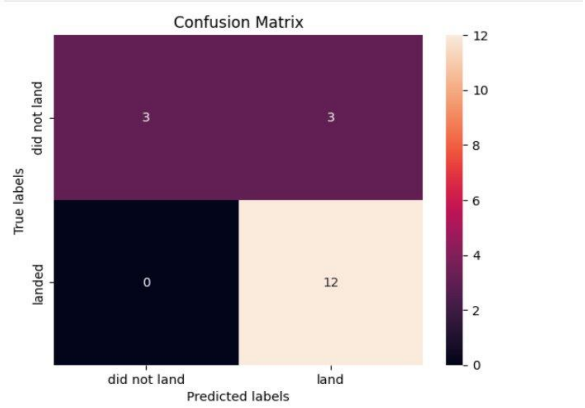
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CONFUSION MATRIX EVALUATION

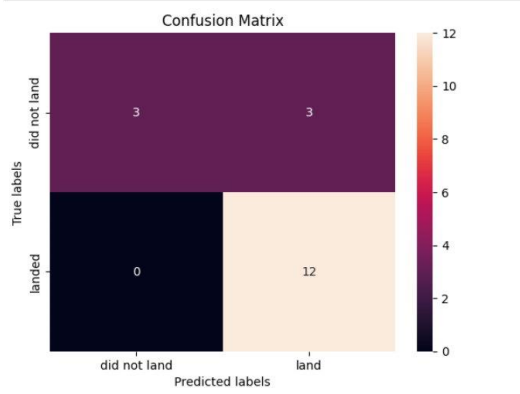
Predictive Analytics

*(Note: All four models achieved an identical test set accuracy of **83.33%** on 18 test samples)*

- **Logistic Regression:**
  - **Test Accuracy:** 83.33%
  - **Confusion Matrix:** True Negatives: 3 | False Positives: 3 | False Negatives: 0 | True Positives: 12



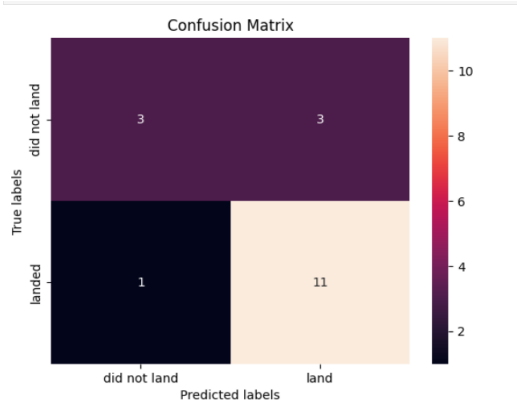
- **Support Vector Machine (SVM):**
  - **Test Accuracy:** 83.33%
  - **Confusion Matrix:** True Negatives: 3 | False Positives: 3 | False Negatives: 0 | True Positives: 12



https://github.com/Geethima-Rajapaksha/ibm-datascience-capstone-project/tree/Machine-Learning-Prediction-lab

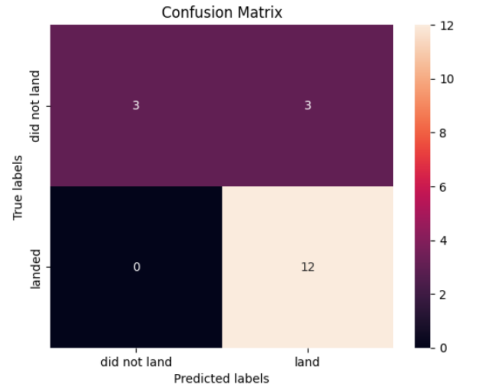
CONFUSION MATRIX EVALUATION

- Logistic Regression: 83.33%
  Decision Tree Classifier:
- Test Accuracy: 83.33%
- Confusion Matrix: True Negatives: 3 | False Positives: 3 | False Negatives: 0 | True Positives: 12



- K-Nearest Neighbors (KNN):
- Test Accuracy: 83.33%
- Confusion Matrix: True Negatives: 3 | False Positives: 3 | False Negatives: 0 | True Positives: 12

K-Nearest Neighbors(KNN): 83.33%



Summary:

- All four tuned classification models achived the same test accuracy (83.33%).
- Logidtic Regression or SVM is recommended for production due to lower computational complexity and better generalizability on binary landing classifications.

CONCLUSION & FINAL TAKEAWAYS

Final Takeaways & Business Value

Conclusion

- **Predictive Performance:** Achieved 83.33% accuracy in predicting first-stage Falcon 9 booster recovery using machine learning models.
- **Key Drivers:** Payload mass (specifically in the 2,000–4,000 kg range) and launch site selection are critical indicators of landing success.
- **Business Value:** Accurate landing predictions allow SpaceX to price launch contracts competitively ($62M vs $165M market baseline) while maintaining operational profitability.

EXPLORATORY DATA ANALYSIS – SUCCESS RATE VS. ORBIT TYPE

Exploratory Data Analysis (EDA)

Key Observations:

- **100% Success Rate Orbits:** Missions targeting **ES-L1**, **GEO**, **HEO**, and **SSO** achieved a 100% landing success rate.
- **Moderate Success Rates: LEO** (~71%), **VLEO** (~85%), and **ISS** (~62%) showed high-to-moderate success rates as launch techniques matured over time.
- **Challenging Orbits: GTO** had a lower success rate (~52%) due to higher payload masses and higher re-entry speeds required for geostationary transfer.
- **Lowest Success Rate: SO** (Polar Orbit) had a 0% success rate during early test launches.

Strategic Insight:

- Orbit selection directly affects the booster's re-entry velocity and remaining fuel, which dictates whether a drone ship or ground pad landing is feasible.